Graph-Structured Convolution-Guided Continuous Context Threshold-Aware Networks for Hyperspectral Image Classification

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***Abstract*— Although convolutional neural networks (CNNs) have shown superior performance to traditional machine learning algorithms for hyperspectral image (HSI) classification tasks, the ability of traditional CNNs to model remote dependencies in the spatial orientation of HSIs is still limited, and they always extract similar low-level features, leading to feature redundancy. To cope with this limitation, this article proposes a novel multiorder statistical representation-guided graph convolution and continuous context threshold-aware network for the classification of HSIs with limited training samples. Initially, the spectral–spatial information is separately modeled using first-order features and second-order pooling operators. Secondly, we graph-structured the patch features, and by employing a random walk transition probability matrix, the graph-structured convolution can mine more discriminative directional features. In addition, a continuous context threshold-aware network is designed to model multidimensional spatial relationships, which enhances the feature representation of graph features. Specifically, the cross-attention mechanism is used to calculate the attention weights in the vertical and horizontal directions, and the features are divided into two levels—important and secondary—by solving the cosine distance between feature vectors; the former is retained and the latter is punished. Extensive experiments on multiple HSI datasets demonstrated that the proposed method delivers competitive performance. The code will be available at https://github.com/vivitsai/GSC-CCTA.**

***Index Terms*— Continuous context threshold-aware network, graph-structured convolution, imbalanced training data, neural networks.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are rich in spatial and spectral data of a host of features and have hundreds of spectral bands within the same location, which can give essential identification information for a variety of ground objects. Each pixel can be understood as a high-dimensional vector, so the value of the pixel corresponds to the spectral reflectance from infrared to visible light. Recently, hyperspectral data collection has become easier, making HSIs a promising technology for numerous applications, including target classification and identification in geological exploration [1], fine agriculture [2], ecology [3], urban remote sensing [4], and military target detection [5], [50].

In HSIs [6], each pixel has hundreds of spectral bands, and this high-dimensional information leads to many challenges in the classification task. The limited labeled samples and high-dimensional imbalance make it difficult to integrate spectral and spatial information to fully utilize their complementary properties. Due to the high computational cost of spectral bands, HSIs' classification focuses on spectral and spatial information extraction.

Dimensionality reduction is one of the most commonly used HSIs' classification strategies from the early studies [7] to reduce the number of dimensions in spectral data. Although dimensionality reduction alleviates the high dimensionality and the computational complexity of hyperspectral data to some extent, these dimensionality reduction methods ignore the spatial correlation and contextual information of the spectral data. Xue et al. [8] propose modeling representative features using attention-based first and second-order pooling operators to capture the spatial–spectral information of HSIs, whose generalization performance, classification accuracy, and convergence speed are better than traditional dimensionality reduction-based methods. Shang et al. [9] came up with an HSIs' classification model using second-order channel attention. The second-order channel attention interacts with channel features via the covariance matrix, and long-term dependencies between channels are found. Therefore, the multiorder statistical representation outperforms traditional dimensionality reduction methods in feature mining, and more importantly, the multiorder statistical representation can preserve the spatial correlation and contextual information of the spectrum.

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In recent years, there has been an increasing amount of work on HSIs' classification using convolutional neural networks (CNNs)-based methods and graph-based approaches. Ahmad et al. [10] proposed to divide the HSIs' cube into small overlapping 3-D blocks in order to generate 3-D convolutional feature maps in a computationally efficient manner using 3-D kernel functions over multiple consecutive bands of spectral information, which improves the accuracy of the classification model. Dong et al. [14] focused on helping classifiers obtain distinguishable spectral–spatial features and proposed a deep ensemble CNNs method based on sample expansion, which achieved excellent classification performance on a small training set. Xu et al. [15] proposed a full convolution network for HSIs' classification and combined spatial and spectral features using adaptively learned weighting factors that achieved a competitive and good performance. Although CNNs perform very well in the spatial–spectral domain, it is difficult to establish long-range dependencies and explicitly extract information from spectral band sequences due to the local connectivity and translation invariance of CNNs. While graph-based methods are expected to alleviate this limitation, Zhang et al. [11] proposed a collaborative propagation method based on dual sparse representation graphs for HSIs' classification, which employs sparse representations to construct spectral and spatial graphs in spectral and spatial dimensions, respectively, and obtains competitive HSIs' classification results. In order to extract more discriminative features, Kong et al. [12] utilized the rich spectral features and spatial context information to construct a more reliable spectral–spatial graph, which reduces the negative impact on the learning process of the prediction model. Jiang et al. [13] constructed affinity graphs based on spectral and spatial similarities, and measured their similarity by spectral feature vectors to transform label noise purification into an optimization problem with graph constraints. Multilayer spectral–spatial graphs based on multiscale segmentation were also developed to provide richer spatial information and significantly improve classification accuracy. There are also many works that combine graphs with CNNs, Guo et al. [47] combined graphs with residual CNNs and proposed a multilevel graph residual module, which can obtain multilevel spatial correlation between neighboring hyperpixels by dividing HSIs' pixels into hyperpixels and serving as input nodes for this module when extracting topological information. Lin et al. [48] combined graphs with U-Net and proposed the context-aware attention graph U-Net, which is capable of extracting intra-class embeddings in non-Euclidean space by combining similarly distributed feature vertices. Liu et al. [49] used CNNs and GCNs to learn features for small-scale regular and large-scale irregular regions, respectively, and generate complementary spectral–spatial features at the pixel level and the superpixel level, respectively. Features are also propagated between image pixels and graph nodes through the use of graph encoders and decoders, thus allowing the CNNs and GCNs to work together in a single network and making the graph more adaptive to HSIs content. The above approach of combining graphs and CNNs actually focuses on the collaborative work of the two and does not generalize CNNs to graph-structured data in general.

Thus, inspired by the work [16] proposed by Hechtlinger et al., this article generalizes CNN to the nongrid-structured data with local connectivity and stability. Different from the standard graph model, it considers the image as a directed graph with edges between adjacent pixels, which can better reflect the dependency between each node and its nearest neighbor. To perform convolution, neighboring pixels of a given pixel are usually selected, and their weights and inner products are computed. A spatial convolution operator that randomly strolls through a graph and picks the first N nearest neighbors of each node is necessary for applying CNN to graph-structured data. After that, the N nearest neighbors are selected for each node based on their positional proximity to the nodes, and their weight and inner product are calculated using convolution.

In light of the aforementioned works, this article proposes a novel graph-structured convolution-guided continuous context threshold-aware network for HSIs' classification of low training samples. Initially, spectral–spatial information is modeled separately using first-order features and second-order pooling operators, instead of using traditional dimensionality reduction methods, such as LDA. Second, we improved the generalization of CNN to graph-structured data by graph-structuring the patch features. By using a random walk transition probability matrix, graph-structured convolution can mine more discriminative directional features. Furthermore, we design a continuous context-threshold-aware network to model multidimensional spatial relationships, thereby enhancing feature representation. In particular, the cross-attention is applied to calculate the weights in the vertical and horizontal directions, and the features are divided into important and secondary levels by solving the cosine distance between feature vectors, with the former being retained and the latter being penalized. Extensive tests on multiple HSI datasets indicate that the designed algorithm achieves competitive results.

The principal contributions include the following.

1) We propose an improved graph-structured convolution in which the graph-structured convolution can mine more discriminative directional features by using a random walk transition probability matrix. We propose to use the square of the correlation matrix as the similarity matrix, which is because the correlation between features can capture the spatial dependence of HSI data, and in addition, utilizing exponential linear unit (elu) as the activation function of graph-structured convolution is advantageous for preserving semantic information in the negative part.
2) A novel continuous context threshold-aware module is designed to model multidimensional spatial relationships. It uses a bidirectional GRU to extract the temporal features of the graph random walk convolution process to enhance the feature representation and uses a cross-attention mechanism to compute the attention weights in the vertical and horizontal directions of the graph random walk convolution to further enhance the local connectivity and directionality within the graph.
3) We also propose a novel threshold-aware strategy to classify the feature maps into two levels, important

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and secondary, by solving the angle cosine distance between the cross-attention weight vectors, retaining the important features, and penalizing the secondary features, which improves the sensitivity of the model to more discriminative features.

4) Extensive testing results on four well-known HSI datasets show that the proposed algorithm achieves competitive results with low training samples, proving its effectiveness and robustness.

II. Related Works

A. CNN-Based Graph Learning

Processing graph-structured data is more complex than processing image data, and the size of graphs is arbitrary, lacking the same spatial locality as images. In recent years, graph learning [17] has been the subject of extensive research, particularly on the basis of CNN. The rise of CNN has become an emerging field to generalizing CNN to graph-structured data. Bronstein et al. [18] outline two primary strategies for generalizing CNN to graph-structured information: spectral methods and spatial methods. To generalize the CNN operator, spectral approaches are based on the feature vectors derived from the spectrum decomposition of the graph Laplacian operator. It is possible to train convolutional layers with many parameters irrespective of the input size for low-dimensional graphs, resulting in efficient deep architectures. Ma et al. [19] offer an enhanced spectral-based GCN for directed graphs by refining their propagation model with the redefined Laplace operator, which permits the analysis of directed graph data in semisupervised node classification. Henaff et al. [20] improve the GCN for directed graphs by latticing the Laplacian operators defined as graph Laplacian operators to perform Fourier transforms, constructing deep architectures with small learning complexity over non-Euclidean domains. However, the main drawback of the spectral model is its dependence on the graph since the graph filters based on the spectral approach are functions of the Laplace determinant of a particular graph and are, therefore, restricted to a fixed graph structure. Despite the fact that Kumar et al. [21] attempted to learn structured graphs from observed samples to combine the spectral features with Gaussian graph modeling, they were unable to do so. Nevertheless, because it is based on the Laplace operator, the algorithm is restricted to a fixed graph structure.

The diffused CNN that Atwood and Towsley [22] proposed is a spatial technique for graph-structured data. It learns diffused representations from graph-structured data and employs them as the foundation for node classification. This spatial method utilizes a random walk on the graph to select spatial neighbors for convolution while preserving shared weights. Bai et al. [23] created a spatial graph convolution network model without backtracking alignment to learn the features of graph classification. The motivation of the spatial method is to convert the graph of arbitrary size into a grid structure with no backtracking alignment of fixed size and carry out graph convolution operation with a new space related to the grid structure, alleviating the problem of information loss and imprecise information representation and further generalizing CNN on graph-structured data. Danel et al. [24] exploit spatial information to efficiently learn from graphs that occur naturally in space. They take into account the order of neighbors of nodes by utilizing node locations and provide a geometric interpretation of network vertices.

Although the spatial method-based graph convolution operator is preferable to the spectral approach, the spatial method is position-dependent, and it is challenging to guarantee translation invariance for nongrid data. Therefore, the spatial convolution operator proposed in this work takes advantage of the relative distance between nodes and uses the elu function to consider the semantic information of the negative part.

B. Graph Learning-Based HSIs Classification

Numerous graph learning-based solutions to the HSIs' classification problem have been presented [13], [25], [26]. Gao et al. [40] constructed a simple graph with vertices representing a pixel and edge weights encoding the similarity between two pixels to establish the proximity between HSIs' pixels from high-dimensional features. Im and Taylor [41] used binary classifiers to learn the neighborhood graph used in semisupervised learning. These binary classifiers were trained to predict whether a pair of HSIs' pixels shares the same label. Ma et al. [42] proposed that sparse graphs using local machine-learning-based methods can capture the local geometric properties of hyperspectral data and, therefore, facilitate the performance of classifiers. Xue et al. [43] achieved good classification performance of HSIs by constructing sparse graphs through matrix computation and using linear graph embeddings to generate low-dimensional features. Xue et al. [44] then introduced a Gaussian kernel based on which can adaptively weighted sparse graph regularization method that can adaptively model the neighborhood information, and the overall accuracy (OA) of HSIs' classification is higher than 90%. Su et al. [45] develop a collaborative node embedding method based on graph cuts considering that node embedding facilitates aggregation of correlations, which enhances the generalization ability of feature representation of HSIs by employing an integration framework based on random subspaces to obtain collaborative feature sets. Subsequently, Su et al. [46] consider correlations in non-Euclidean spaces and propose a normalized spectral clustering method, which builds vertices of undirected graphs by a kernel-based iterative filter that assigns initial connections to nodes associated with pixels and is able to not only collect local correlations in Euclidean spaces but also capture global correlations in manifolds, achieving competitive performance on the HSIs' classification task. While Chen et al. [35] optimized the HSIs' classification task by adaptively learning the data graph in a subspace to capture the local data stream shape and extracting the spatial information in the spectral images with regularization terms. By averaging the distance between unlabeled and labeled samples for each category, Luo et al. [27] developed a semisupervised graph model. A semisupervised graph model is created by ranking unlabeled data and connecting them to all labeled samples from their nearest neighbor classes, which maximizes category differentiation and preserves local neighborhood information. The graph learning strategy presented

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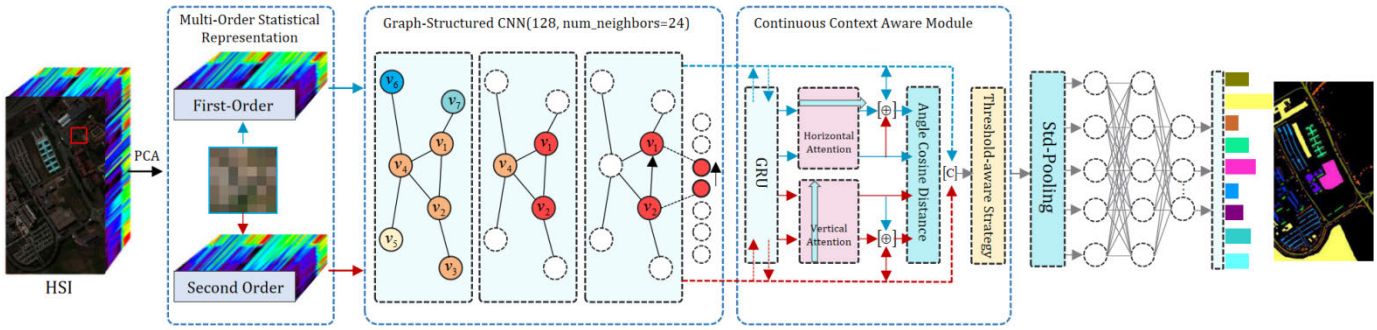


Fig. 1. Schematic of the overall construction of the GSC-CCTA model. ⊕ represents the elementwise addition, and C represents the feature concatenation.

by Sellars et al. [28] uses a superpixel technique to determine meaningful local regions in HSIs, given that these regions may share the same categorical labels. The recovered spectral and spatial information is then utilized to build systolic weighted graph representations, where each node represents an area as opposed to a pixel, with classification results that beat the CNN-based approaches compared. Thus, graph learning can aggregate features of neighboring nodes in non-Euclidean space, thus establishing remote dependencies of different nodes in the space, which can significantly improve the classification performance of HSIs.

In other studies, attention-based graph learning has also been investigated, and Wan et al. [29] apply an attention weight to adaptively describe the spatial correlation between image regions. The produced local feature representations can subsequently be used to encode global contextual information. Ultimately, predictions can be made using local and global contextual knowledge. Ding et al. [30] suggested a classification model for HSIs based on samples of accumulated attention and graphs. Inspired by the attention mechanism, it uses the graph attention mechanism to describe the relevance between neighboring regions of space, allowing it to automatically acquire the graph's deep context knowledge by focusing on significant spaces. Xu et al. [31] created a spectral–spatial residual graph attention network that can adaptively aggregate local spatial information from neighboring nodes using the graph attention. The performance of the aforementioned graph learning methods on HSI data is superior to that of the benchmark graph learning methods, demonstrating that attention mechanism is also possible in graph learning. However, the existing methods only consider attention weight from a single direction, which may lead to suboptimal results. Therefore, this article calculates the weight of attention from the horizontal and vertical directions, and extracts the discriminative features by solving the angle cosine distance between the two groups of attention weight vectors.

Unlike these traditional graph learning-based approaches, the focus of this article is to build deeper and more generalized networks using fewer training samples based on the original distribution of data samples. Therefore, we not only improve the graph-structured convolution but also enhance the feature representation by extracting temporal features of the graph random walk convolution process through bidirectional GRU and further enhance the local connectivity and directionality in the graph using a cross-attention module to compute the weights of the horizontal and vertical directions of the graph random walk convolution.

III. Methodology

The proposed algorithm focuses on the HSI classification with few training samples, for which we propose a novel graph-structured convolution-guided continuous context threshold-aware model. First, first- and second-order pooling operators have been applied to model spectral–spatial information, respectively. Secondly, we graph-structured the patch features, and by employing a random walk transition probability matrix, the graph-structured convolution can mine more discriminative directional features. In addition, a continuous context threshold-aware network is designed to model multidimensional spatial relationships, which enhances the feature representation. Specifically, the cross-attention mechanism is utilized to calculate the weights in the vertical and horizontal directions, and the features are separated into two levels of important and secondary by calculating the cosine distance between feature vectors, with the former enhancing and punishing the latter. The schematic of the GSC-CCTA can be seen in Fig. 1.

A. Multiorder Statistical Representation

1) First-Order Statistical Feature: To gain more discriminative low-dimensional semantic information and save computational cost, the proposed model does not simply use PCA to lessen the dimensionality of HSIs but is also inspired by the A-SPN model [8]; a square moving window ($h \times h$) is used to generate patch features reflecting spatial–spectral information; and the rectified linear unit (ReLU) function is employed to limit the risk of overfitting. Let $X \in \mathrm{R}^{\mathrm{N}\times\mathrm{B}}$ be an HSI with B bands and C classes; we define the window $w_i^{h\times h}$ as

$$w_i^{h\times h} = \{x_{i-h}, \ldots, x_i, \ldots, x_{i+h}\}. \tag{1}$$

The parameters h are nonnegative numbers that determine the number of samples to include prior to and subsequent to the sample x_i. The first-order statistical feature representation for the window $w_i^{h\times h}$ can be computed, and then, the window is shifted forward by one sample to focus on x_{i+1}. The total amount of samples in the window is $K = h^2$. When processing samples near the end of the input, if there are not enough samples to fill the window defined above, zeros should be

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inserted to build a full window of length K. Then, the input can be represented as:

$$\tilde{x} = \left\{ \underbrace{0, \ldots, 0}_{h \text{ zeros}}, x_1, x_2, \ldots, x_{n-1}, x_n, \underbrace{0, \ldots, 0}_{h \text{ zeros}} \right\} \tag{2}$$

where $\tilde{x}$ represents a sample set of a window input to the first-order statistical operator and n represents the total number of nonzero samples of a window. Finally, batch normalization is utilized to confirm the input data X to the same distribution to expedite the network's convergence speed and prevent gradient expansion and gradient disappearance. Therefore, the first-order statistical representation of the unit moving window learned from HSIs can be calculated by the following equation:

$$F_m^i = \mathrm{BN}\left(\mathrm{ReLU}\left(w_i^{h\times h}\right)\right), \quad 1 \le i \le \frac{N \times B}{K} \tag{3}$$

where F_m^i represents the first-order statistical representation of the unit-squared moving window, ReLU denotes the linear rectification function, N denotes a total of N pixels, and BN represents the batch normalization operation.

2) Second-Order Statistical Feature: To further obtain a second-order statistical representation of the first-order features, the proposed algorithm proposes to use a second-order pooling operator [8] in the form

$$F_s^i = \left(F_m^i\right)^T \circ F_m^i \tag{4}$$

where F_s^i represents a symmetric matrix and $\circ$ represents the inner product operation.

The second-order pooling operator focuses on the correlation between spectra at the same spatial location, makes full use of the HSIs information, and can aggregate the second-order statistics of the block features, which is conducive to capturing the global spatial–spectral information.

B. Graph-Structured Convolution

Standard CNNs implicitly utilize the basic property of the input and can only use the input of grid structure because it learns the local rectangular filter through repeated convolution on multiple blocks, which makes the local region highly correlated, while the global region weakly correlated. Inspired by Hechtlinger et al., to effectively mine the global correlation of independent spatial positions of HSIs, this article introduces an improved graph-structured CNN. In contrast to the standard convolution operator, graph-structured CNN can do a random walk on the graph. To choose the first n nearest neighbors for each node, and then, for each node, the weight and inner product of the selected n nearest neighbors for the convolution calculation are sorted by their relative locations to the node. As shown in Fig. 2, this enables us to convolution with the same shared weights on each node and to mine the directed dependencies between each node and its nearest neighbor. The graph-structured convolution operation is equivalent to the standard convolution operation when the edges between adjacent pixels of the image are considered undirected.

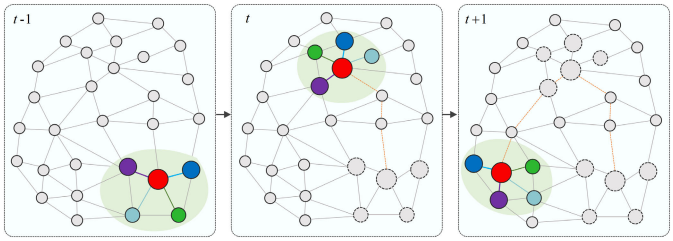


Fig. 2. Diagrammatic representation of graph-structured convolution with a size of 5 and a stride of 3. For a given node, the convolution process is carried out by the node and its four random walk-selected nearest neighbors. In addition, the random walk can be expanded to generate higher dimensional neighbors, with the weight of the graph-structured convolution operation being proportional to the distance between the node and the neighbor.

1) Transition Probability Matrix: In this study, a graph transition probability matrix is utilized to determine the expected number of visits for a random walk beginning at a node, and convolution operations are conducted on the top n nodes with the highest expected number of visits to properly select their immediate neighbors effectively. For convolutions on a single graph of a single layer, we let $G = \{\upsilon, \varepsilon\}$, where $\upsilon = \{v_1, v_2, \ldots, v_N\}$ is the set of nodes of size $N = |\upsilon|$ and $\varepsilon = \{e_1, e_2, \ldots, e_M\}$ is the set of edges of size M. If T represents a random walk probability matrix on the graph, then T_{ij} indicates the probability of traveling from node v_i to node v_j. If we let S represent the similarity matrix of the graph and C represent the correlation matrix of the graph, then we can construct the diagonal matrix D using the following equation:

$$D_{ii} = \sum_j S_{ij}. \tag{5}$$

If the data graph structure is known, the graph probability transition matrix can be denoted as

$$T = D^{-1}S \tag{6}$$

where T represents a random walk probability matrix on the graph.

Since the correlation between features captures the geometric structure of the HSI, this work proposes using its square as the similarity matrix if the data graph structure is unknown. The equation for generating the similarity matrix is given as follows:

$$S_{ij} = (C_{ij})^2 \quad \forall i, j \tag{7}$$

where C represents the correlation matrix of the graph.

2) Node Visits: After obtaining the transition probability matrix T, the number of visits to a given target node can be calculated. Let T_{ij} be the transition probability from node v_i to v_j in k steps; then,

$$\begin{cases} H^{(0)} = E \\ H^{(1)} = E + T \\ \quad \vdots \\ H^{(k)} = \sum_{i=1}^{k} \left(\prod_{i=1}^{k} \left(T^{(k)}\right) + E \right) \end{cases} \tag{8}$$

where $H^{(k)}$ represents the frequency of visiting v_j in k steps starting from v_i, and i and j represent the index of the node.

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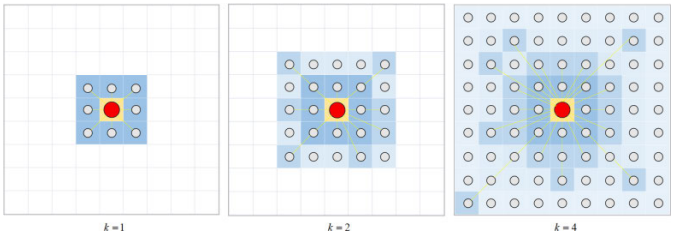


Fig. 3. Schematic visualization of a row $H^{(k)}$ of a graph in 2-D space. Each node is connected to eight adjacent nodes. If $k = 1$, most of the weight is concentrated on the node, and the weight of the first-order neighbors is small, which is equivalent to a traditional 3×3 convolution. If the step size $k \geq 2$, it can provide more shared weights for further neighbors without losing local semantic information.

H in row i computes the similarity matrix between node v_i and its neighbors through a random walk of the graph, which further merges neighbors further away from the node as step k increases, while assigning compatible weights to the node and its nearest neighbors through a summation operation. Fig. 3 provides a visualization of the matrix H on a 2-D space.

3) Graph-Structured Convolution Operator: Each line of $H^{(k)}$ can be applied to the node to determine its nearest neighbor, so $H^{(k)}$ of the ith line can be used to perform convolution operation on the graph node v_i. We first sort the $H^{(k)}$ sequence in ascending order. Then, there are the following inequalities:

$$H_{i,N}^{(k)} < H_{i,N-1}^{(k)} < H_{i,N-2}^{(k)} < \cdots H_{i,1}^{(k)}. \tag{9}$$

We define GSC as a convolution operation of size q on a graph G with nodes $\upsilon = \{v_1, v_2, \ldots, v_N\}$ and the nearest q neighbors of each node and the weight w as the inner product; then, the calculation equation of the graph-structured convolution operator is given as follows:

$$\text{elu} = \begin{cases} C_{i,j}, & C_{i,j} > 0 \\ \alpha(e^{C_{i,j}} - 1), & C_{i,j} \leq 0 \end{cases} \tag{10}$$

$$g_{i,j}^{(k)} = \text{elu}(C_{i,j}) H_{i,j}^{(k)} v_j \tag{11}$$

$$\text{GSC}(\upsilon) = \begin{bmatrix} g_{1,1}^{(k)} & g_{1,2}^{(k)} & \cdots & g_{1,q}^{(k)} \\ g_{2,1}^{(k)} & g_{2,2}^{(k)} & \cdots & g_{2,q}^{(k)} \\ \vdots & \vdots & \ddots & \vdots \\ g_{N,1}^{(k)} & g_{N,2}^{(k)} & \cdots & g_{N,q}^{(k)} \end{bmatrix} \bullet \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_q \end{bmatrix} \tag{12}$$

where

$$\upsilon = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}$$

and $\bullet$ represents the inner product operation. The variable with the highest value in each column of the matrix $g^{(k)}$ will be convolved with w_i, which also means that the weights w_i will be constrained by the decision distance of the probability transition matrix. We use elu as the activation function of graph-structured convolution to help preserve the semantic information of the negative part.

4) Power of Node Visits: The choice of k values is related to the HSI data. First, k must be large enough to cover the first n neighbors of each node. If the transfer matrix T is sparse, it needs a higher value of k. Second, by expressing π as a Markov chain stationary distribution according to the properties of stochastic process, we can get

$$\lim_{k \to \infty} \frac{H_{ij}^{(k)}}{k} = \pi_j \quad \forall i, j \tag{13}$$

where $H_{ij}^{(k)}$ represents the frequency of visiting v_j in k steps starting from v_i.

From the above equation, it can be seen that, if the value of k is large, the local information will be smoothed, so the value of k should be kept relatively low as much as possible, and after experimental verification, k is set to 4.

C. Continuous Context Threshold-Aware Module

To further increase the representational capability of graph-structured convolutional features, this article also proposes a novel continuous context threshold-aware module. The temporal features of the graph random walk convolution are first extracted using a bidirectional GRU, followed by the calculation of attention weights in horizontal and vertical directions using a cross-attention mechanism; then, the feature maps are classified into two levels, important and secondary, by solving for the cosine distance between the attention weight vectors, the important features are enhanced, and the secondary features are penalized.

1) Graph-Structured Timing Feature Extraction: Graph-structured convolution mines the high-dimensional semantic information with an intrinsic correlation of input data. This work suggests using bidirectional GRU for feature extraction to further extract the temporal features of graph-structured convolution random walk. In practice, RNNs have difficulty identifying the dependencies of long time steps in time series. The gated recurrent unit (GRU), which is controlled by gates that may learn data flow, aims to improve this situation. In the GRU, the current input F_{GSC} represents the output of the graph-structured convolution operator. GRU assists in the calculation of subsequent hidden states by computing candidate hidden states. By calculating the Hadamard product between the yield of the reset gate at the present time step and the previous time step's hidden state, we may determine whether or not the corresponding element in the hidden state is zero. If the reset gate element value is near 1, the previous time step's hidden state is kept, the Hadamard product is added to the input for the present time step, and the candidate hidden state h_{t-1} is computed using the fully connected (FC) layer and its calculation equation can be described as follows:

$$r_t = \sigma\left(F_t^{\text{GSC}} W_{xr} + h_{t-1} W_{hr} + b_r\right) \tag{14}$$

$$u_t = \sigma\left(F_t^{\text{GSC}} W_{xz} + h_{t-1} W_{hz} + b_z\right) \tag{15}$$

$$\tilde{h}_t = \tanh(x_t W_{xh} + (r_t \bullet h_{t-1}) W_{hh} + b_h) \tag{16}$$

$$h_t = u_t \bullet h_{t-1} + (1 - u_t) \bullet \tilde{h}_t \tag{17}$$

where $h_t \in R^{n \times h}$ represents the hidden state. u_t denotes the output of the update gate. $\tilde{h}_t$ and r_t denote the candidate hidden

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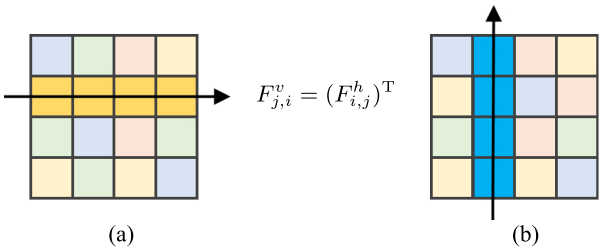


Fig. 4. Schematic of the structure of the cross-attention. (a) Horizontal direction. (b) Vertical direction.

state and output of the reset gate of the GRU at time step t, respectively.

Bidirectional GRU can better capture the deep semantic features of graph random walk convolution by combining forward and backward GRUs.

2) Cross-Attention: As depicted in Fig. 4, the proposed cross-attention yields weights from vertical and horizontal directions. Let the structured graph convolution output the feature map as $g^h_{i,j} \in R^{H\times W}$, where $H \times W$ denotes the size of the feature map, and $g^k_{i,j}$ is input to the horizontal attention unit to calculate the weight in the horizontal direction and its calculation formula can be defined as follows:

$$a_h = \frac{e^{(W_h g_{i,j} + b_h)}}{\sum_{i,j} e^{(W_h g_{i,j} + b_h)}} \tag{18}$$

where W_h and b_h are the weight parameters of the dense layer and a_h is the coefficient along the horizontal attention.

The weight in the vertical direction is calculated by inverting the feature matrix, and the process is expressed by the following equation:

$$g^v_{j,i} = \left(g^v_{i,j}\right)^T \tag{19}$$

where $g^v_{j,i}$ indicates the feature map that has been transposed. Therefore, we can obtain attention weights in the vertical direction in the same way, whose formula is declared as follows:

$$a_v = \frac{e^{(W_v g_{j,i} + b_v)}}{\sum_{j,i} e^{(W_v g_{j,i} + b_v)}} \tag{20}$$

where W_v and b_v are the weight parameters of the dense layer and a_v is the coefficient along the vertical attention.

3) Angle Cosine Distance: This article proposes to use the angle cosine distance to calculate the similarity of the cross-attention weight vectors, as shown in Fig. 5. Since the degree to which the two sets of cross-attention weights are similar can be interpreted as a reflection of how easily the acquired features can be differentiated from one another, this article makes this suggestion. This indicates that the feature of the current attention weight mapping is more discriminative since the cosine value is close to 1, and the angle tends to zero. The fact that the cosine value is so close to zero and the angle is approaching ninety degrees is a strong indication that the feature may be superfluous. $\cos\theta$ represents the similarity between attention weights a_h and a_v, independent of their norms, so the calculation equation of the cosine distance

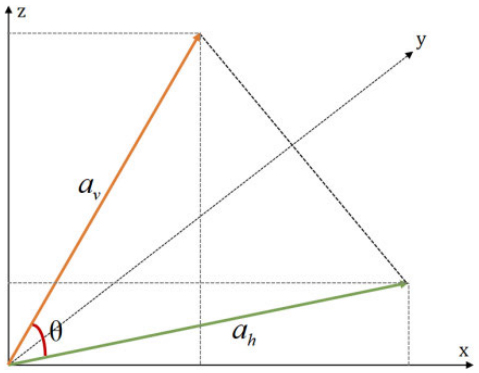


Fig. 5. Schematic of the angle cosine distance.

between the two sets of cross-attention weights is given as follows:

$$\cos\theta = \frac{\sum_{i=1}^{n} a_i^{(h)} a_i^{(v)}}{\sqrt{\sum_{i=1}^{n}\left(a_i^{(h)}\right)^2}\sqrt{\sum_{i=1}^{n}\left(a_i^{(v)}\right)^2}} \tag{21}$$

where $a_i^{(h)}$ and $a_i^{(v)}$ represent the components of two sets of cross-attention weight vectors a_h and a_v, respectively.

4) Threshold-Aware Strategy: After obtaining the angle cosine distance, this article further proposes a novel threshold-aware strategy for feature screening; specifically, when the threshold value is lower than the thr, we consider the angle cosine distance to be larger, indicating that the semantic information obtained from feature points in different directions is not highly similar; and when the threshold value is higher than the thr, we consider the semantic information obtained from feature points in different directions to have higher similarity so that the sparsity of the feature matrix can be increased to achieve better feature screening effect. Feature screening is achieved by enhancing and penalizing the features, that is, features above the threshold receive a value of 1 to enhance it, and features below the threshold receive a value of 0 to penalize it. The calculation equation of the threshold-aware strategy can be expressed as follows:

$$\begin{cases} \mathrm{TA}_h(\zeta) = \begin{cases} \mathbf{0}, & \cos_h \le \mathrm{thr} \\ \mathbf{1}, & \cos_h > \mathrm{thr} \end{cases} \\ \mathrm{TA}_v(\zeta) = \begin{cases} \mathbf{0}, & \cos_v \le \mathrm{thr} \\ \mathbf{1}, & \cos_v > \mathrm{thr} \end{cases} \end{cases} \tag{22}$$

where $\mathrm{TA}_h()$ and $\mathrm{TA}_v()$ denote the threshold-aware strategy function in the vertical and horizontal dimensions, and thr denotes the threshold.

From (21), we can see that, when useful features need to be retained, we convert all the weight coefficients of features identified as important levels to 1. When useless features need to be eliminated, we convert the weight coefficients of features identified as secondary level to 0. The sensitivity of the threshold-aware strategy to more discriminative features further improves the representation capability of the model and also further screens the temporal features of the bidirectional graph random walk convolution. The output of a continuous

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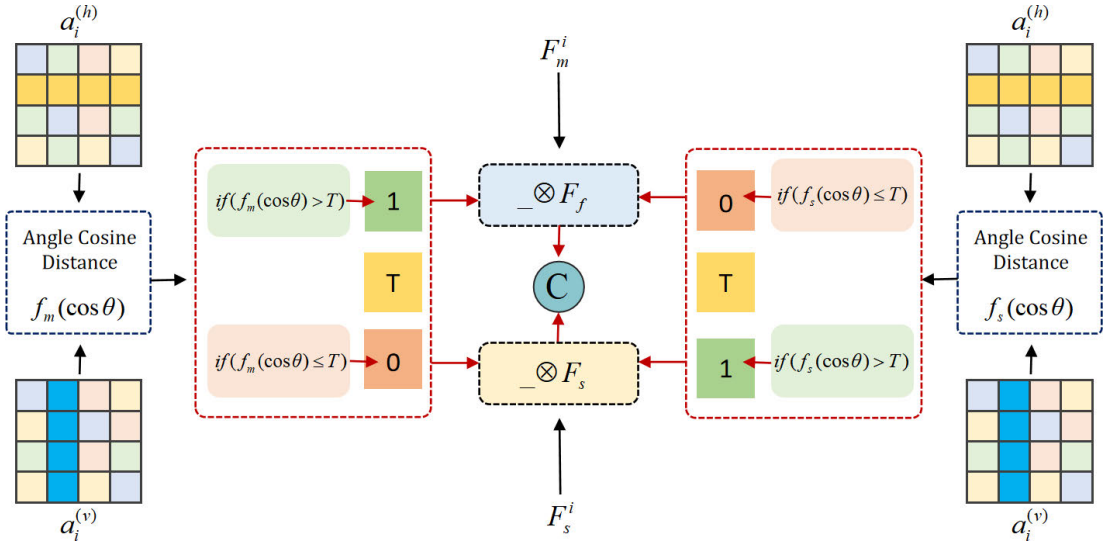


Fig. 6. Schematic of the threshold-aware strategy based on the angle cosine distance.

context threshold-aware module can be defined as

$$F_{\mathrm{CCTA}} = \zeta\left[\mathrm{TA}_h(\cdot)\otimes x_f, \mathrm{TA}_v(\cdot)\otimes x_s\right] \tag{23}$$

where F_{CCTA} represents the output of the continuous context threshold-aware module, ζ represents the feature splicing operation, and $\otimes$ represents the matrix multiplication operator. In addition, Fig. 6 also shows the feature-aware process under the angle cosine distance strategy.

D. Feature Fusion

The features output by graph-structured convolution can further improve the OA of HSIs' classification via feature fusion; we spliced them with the output features of continuous context threshold-aware to obtain the fused features F_U. This process can be described as follows:

$$F_G = \zeta[F_{\mathrm{GSC1}}, F_{\mathrm{GSC2}}] \tag{24}$$

$$F_U = \zeta[F_{\mathrm{CCTA}}, F_G] \tag{25}$$

where ζ represents the feature concatenation operation. F_{GSC1} and F_{GSC2} denote the output features of graph-structured convolution of first-order statistical representation and multiorder statistical representation, respectively, F_G denotes the fusion of two sets of graph-structured convolution features, and F_U denotes the fusion of two sets of graph-structured convolution features and the output features of continuous context threshold-aware module.

E. Classification

In this work, standard deviation pooling is applied to vectorize the output features to obtain a more discriminative low-dimensional representation, which further improves computational efficiency. In addition, standard deviation pooling also plays the role of a flattened layer. The equation for determining the standard deviation is given as follows:

$$\mathrm{Std}(\cdot) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(F_U^i - \mu\right)^2}, \quad \text{where } \mu = \frac{1}{N}\sum_{i=1}^{N} F_U^i \tag{26}$$

$$F_{\mathrm{final}} = \mathrm{Std}(F_U) \tag{27}$$

where $\mathrm{Std}(\cdot)$ represents the standard deviation operation.

The results of standard deviation pooling are calculated based on the variance of all responses within the averaging window. Therefore, standard deviation pooling has a better ability to capture local variability among the responses within each pooling window. Finally, an FC layer with a softmax loss function is implemented to classify HSIs.

IV. Experiments

A. Hyperspectral Datasets

In this article, the proposed algorithm is investigated on four HSI classification tasks involving unbalanced data and small sample data, in which our model is trained with a very small amount of data by limiting the size of the training samples.

1) The Indian Pines (IP) dataset is HSIs gathered in 1992 utilizing the AVIRIS hyperspectral sensor to image an Indian pine tree in Indiana, USA. It comprises 145 × 145 pixels and 224 spectral reflection bands. After removing the spectral bands encompassing the water-absorbed region, 200 spectral bands remain, and the ground truth (GT) consists of 16 classes. Therefore, it is more prone to mixed image elements, which makes classification difficult.
2) The Pavia University (PU) dataset is a portion of the HSI dataset imaged by the German ROSIS hyperspectral sensor in 2003 in the city of Pavia, Italy. The sensor has a number of 115 bands, 12 of which were removed due to noise, resulting in 103 bands of usable data with a spatial resolution of 1.3 m. The GT contains nine categories. The dataset is characterized by complex

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TABLE I
SELECTED TRAINING SAMPLES FOR EVALUATION FROM THE IP DATASET

No.	Name	Total	5%	10%	15%
C_1	Class_Corn-notill	1428	71	142	214
C_2	Class_Alfalfa	46	2	4	6
C_3	Class_Corn	237	11	23	35
C_4	Class_Corn-mintill	830	41	83	124
C_5	Class_Grass-trees	730	36	73	109
C_6	Class_Grass-pasture	483	24	48	72
C_7	Class_Hay-windrowed	478	23	47	71
C_8	Class_Grass-pasture-mowed	28	1	2	4
C_9	Class_Soybean-notill	972	48	97	145
C_{10}	Class_Oats	20	1	2	3
C_{11}	Class_Soybean-clean	593	29	59	88
C_{12}	Class_Soybean-mintill	2455	122	245	368
C_{13}	Class_Woods	1265	63	126	189
C_{14}	Class_Wheat	205	10	20	30
C_{15}	Class_Stone-Steel-Towers	93	4	9	13
C_{16}	Class_Buildings-Grass-Trees-Drives	386	19	38	57

TABLE II
SELECTED TRAINING SAMPLES FOR EVALUATION FROM THE PU DATASET

No.	Name	Total	1%	5%	10%
C_1	Class_Meadows	18649	186	932	1864
C_2	Class_Asphalt	6631	66	331	633
C_3	Class_Trees	3064	30	153	306
C_4	Class_Gravel	2099	20	104	209
C_5	Class_Bare Soil	5029	50	251	502
C_6	Class_Painted metal sheets	1345	13	67	134
C_7	Class_Self-Blocking Bricks	3682	36	184	368
C_8	Class_Shadows	947	9	47	94
C_9	Class_Bitumen	1330	13	66	133

background pixels, a small connected area, and the presence of a large number of discontinuous pixels.

3) The Salinas (SA) dataset was also imaged by the AVIRIS hyperspectral sensor for the Salinas Valley in California, USA. This dataset is less difficult to classify compared to the other three HSI datasets.
4) The Houston dataset was gathered by the CASI-1500 hyperspectral imaging sensor over the University of Houston and its surrounding regions in 2013 with a spatial resolution of 2.5 m, a spectral wavelength range of 380–1050 nm, and 144 bands, making it a large and complex hyperspectral dataset.

B. Experimental Setup

In this work, we propose to evaluate the proposed GSC-CCTA using 0.1%, 1%, 5%, 10%, and 15% training samples on four well-known HSI datasets, as shown in Tables I–IV and Fig. 7. The nine methods that we compared are classified into CNN-based models: SSUN [32], MCNN-CP [33], and Tri-CNN [34]; attention-based models: MDN-ASP [35], RIAN [36], and DMuCA [37]; and graph learning-based models: MDGCN [38], Gia-CFSL [6], and WFCG [39]. For a fair comparison, these nine algorithms have the same parameter settings and training environment as the proposed GSC-CCTA. In addition, the step size k of node access is set to 4, the size of the square moving window of the first-order statistical operator is set to 15×15, the learning rate is $1e-3$, the exponential learning rate decay is $1e-3$, the batch size is 64, the number of iterations is 300, and the threshold T is 0.7.

TABLE III
SELECTED TRAINING SAMPLES FOR EVALUATION FROM THE SA DATASET

No.	Name	Total	1%	5%	10%
C_1	Class_Brocoli-green-weeds-2	3726	37	186	372
C_2	Class_Brocoli-green-weeds-1	2009	20	100	200
C_3	Class_Fallow-rough-plow	1394	13	69	139
C_4	Class_Fallow	1976	19	98	197
C_5	Class_Stubble	3959	39	197	395
C_6	Class_Fallow-smooth	2678	26	133	267
C_7	Class_Grapes-untrained	11271	112	563	1127
C_8	Class_Celery	3579	35	178	357
C_9	Class_Corn-senesced-green-weeds	3278	32	163	327
C_{10}	Class_Soil-vinyard-develop	6203	62	310	620
C_{11}	Class_Lettuce-romaine-5wk	1927	19	96	192
C_{12}	Class_Lettuce-romaine-4wk	1068	10	53	106
C_{13}	Class_Lettuce-romaine-7wk	1070	10	53	107
C_{14}	Class_Lettuce-romaine-6wk	916	9	45	91
C_{15}	Class_Vinyard-vertical-trellis	1807	18	90	180
C_{16}	Class_Vinyard-untrained	7268	72	363	726

TABLE IV
SELECTED TRAINING SAMPLES FOR EVALUATION FROM THE HOUSTON DATASET

No.	Name	Total	0.1%	0.5%	1%
C_1	Class_Stressed grass	1254	13	63	125
C_2	Class_Healthy grass	1251	13	63	125
C_3	Class_Trees	1244	12	62	124
C_4	Class_Synthetic grass	697	7	35	70
C_5	Class_Water	325	3	16	33
C_6	Class_Soil	1242	12	62	124
C_7	Class_Commercial	1244	12	62	124
C_8	Class_Residential	1268	13	63	127
C_9	Class_Highway	1227	12	61	123
C_{10}	Class_Road	1252	13	63	125
C_{11}	Class_Parking Lot1	1233	12	62	123
C_{12}	Class_Railway	1235	12	62	124
C_{13}	Class_Tennis court	428	4	21	43
C_{14}	Class_Running track	660	7	33	66
C_{15}	Class_Parking Lot2	469	5	23	47

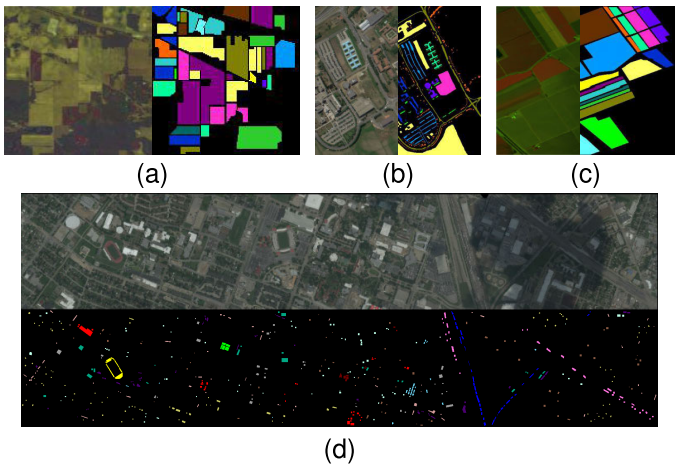


Fig. 7. GT and false-color composite image for each dataset. (a) IP. (b) PU. (c) SA. (d) Houston.

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TABLE V
CLASSIFICATION ACCURACY OF GSC-CCTA ON FOUR DATASETS

IP(5%)		PU(1%)		SA(1%)		Houston(0.1%)	
Class	GSC-CCTA	Class	GSC-CCTA	Class	GSC-CCTA	Class	GSC-CCTA
Alfalfa	0.9318	Asphalt	0.8785	Brocoli_green_weeds_1	0.1000	Healthy grass	0.8634
Corn-notill	0.9719	Meadows	0.9954	Brocoli_green_weeds_2	0.1000	Stressed grass	0.9226
Corn-mintill	0.9531	Gravel	0.8845	Fallow	0.1000	Synthetic grass	0.9970
Corn	0.9424	Trees	0.8843	Fallow_rough_plow	0.9666	Trees	0.9780
Grass-pasture	0.9564	Painted metal sheets	0.9962	Fallow_smooth	0.1000	Soil	0.9617
Grass-trees	0.9654	Bare Soil	0.9977	Stubble	0.9931	Water	0.7639
Grass-pasture-mowed	0.1000	Bitumen	0.9544	Celery	0.9988	Residential	0.8183
Hay-windrowed	0.1000	Self-Blocking Bricks	0.9424	Grapes_untrained	0.9686	Commercial	0.5892
Oats	0.1000	Shadows	0.7846	Soil_vinyard_develop	0.1000	Road	0.8038
Soybean-notill	0.9783			Corn_senesced_green_weeds	0.1000	Highway	0.8954
Soybean-mintill	0.9541			Lettuce_romaine_4wk	0.9971	Railway	0.7293
Soybean-clean	0.9716			Lettuce_romaine_5wk	0.9921	Parking Lot1	0.8501
Wheat	0.9538			Lettuce_romaine_6wk	0.9470	Parking Lot2	0.7683
Woods	0.9958			Lettuce_romaine_7wk	0.9962	Tennis court	0.9692
Buildings-Grass-Trees-Drives	0.9918			Vinyard_untrained	0.9780	Running track	0.1000
Stone-Steel-Towers	0.7191			Vinyard_vertical_trellis	0.1000		
OA	**0.9671**	**OA**	**0.9537**	**OA**	**0.9877**	**OA**	**0.8551**
AA	**0.9553**	**AA**	**0.9242**	**AA**	**0.9898**	**AA**	**0.8607**
Kappa	**0.9626**	**Kappa**	**0.9387**	**Kappa**	**0.9863**	**Kappa**	**0.8433**

C. Evaluation Metrics

To evaluate the robustness of all algorithms in a comprehensive manner, three commonly used metrics, such as OA, which assesses the ratio of correctly classified pixels, kappa coefficient (Kappa), which measures the classification effectiveness by a series of criteria, and average accuracy (AA), which assesses the ratio of correctly classified pixels in each category of features, are used in the experiments to record and evaluate the classification ability of all algorithms

$$f_{\mathrm{OA}} = \frac{1}{N}\sum_{i=1}^{K} p_{ii} \tag{28}$$

where K and N indicate the amount of object classes and the full amount of pixels and p_{ii} symbolizes the element at coordinate (i, i) in the confusion matrix.

AA is calculated by calculating the ratio between the correct prediction of each class and the pixels belonging to this class in the real situation and then taking the average value of the accuracy of each class. The calculation equation is given as follows:

$$f_{\mathrm{AA}} = \frac{1}{K}\sum_{i=1}^{K} \frac{p_{ii}}{\sum_{j=1}^{K} p_{ij}} \tag{29}$$

where $\sum_{j=1}^{K} p_{ij}$ represents the sum of the ith row elements.

The calculation of the Kappa coefficient can be expressed as follows:

$$f_{\mathrm{Kappa}} = \frac{p_0 - p_e}{1 - p_e} \tag{30}$$

where $p_0 = f_{\mathrm{OA}}$ and p_e can be defined as follows:

$$p_e = \sum_{i=1}^{K} \frac{\sum_{j=1}^{K} p_{ij} \cdot \sum_{j=1}^{K} p_{ji}}{N \times N} \tag{31}$$

where $\sum_{j=1}^{K} p_{ji}$ represents the sum of elements in column i.

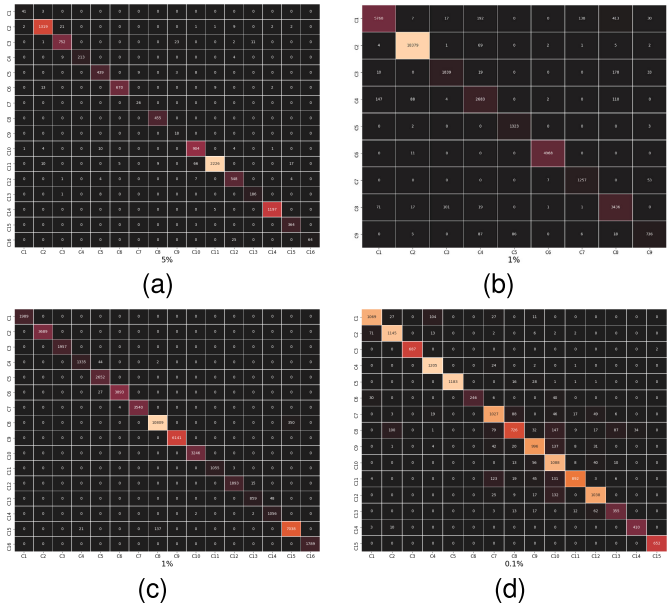


Fig. 8. Confusion matrix. (a) IP. (b) PU. (c) SA. (d) Houston.

D. Comparison Results of Different SOTA Models

From Table V, it can be known that the proposed GSC-CCTA method achieves SOTA performance on all four well-known datasets, especially on the Houston dataset where the OA using only 0.1% of the training samples reaches 85.51%, and the classification performance in each class is just proving the effectiveness of the proposed method. On the IP dataset, the accuracy of all the categories except the class Stone-Steel-Towers reaches more than 93%. The lower accuracy of the class Stone-Steel-Towers is caused by the dispersed and distant locations of this class and the insufficient step size of the graph-structured convolution node visits. The same is true for PU's Shadows and Houston's Water, Parking Lot 2.

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TABLE VI
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON IP

Models	5%			10%			15%		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
SSUN(2018) [32]	0.7379	0.7359	0.7006	0.7782	0.7343	0.7459	0.8563	0.8663	0.8439
MCNN-CP(2020) [33]	0.7769	0.7778	0.7455	0.8356	0.8158	0.8126	0.8596	0.8495	0.8404
Tri-CNN(2023) [34]	0.8926	0.9036	0.8866	0.9146	0.9069	0.9034	0.9245	0.9169	0.9258
MDN-ASP(2021) [35]	0.9038	0.9154	0.9122	0.9274	0.9245	0.9144	0.9425	0.9397	0.9547
RIAN(2022) [36]	0.9199	0.9187	0.9131	0.9285	0.9301	0.9241	0.9431	0.9411	0.9550
DMuCA(2022) [37]	0.9233	0.9310	0.9289	0.9693	0.9597	0.9650	0.9569	0.9608	0.9587
MDGCN(2019) [38]	0.9236	0.9318	0.9068	0.9686	0.9580	0.9642	0.9726	0.9679	0.9739
Gia-CFSL(2022) [6]	0.9517	0.9389	0.9479	0.9715	0.9602	0.9734	0.9827	0.9811	0.9767
WFCG(2022) [39]	0.9604	0.9429	0.9496	0.9786	0.9651	0.9799	0.9903	0.9815	0.9837
GSC-CCTA	**0.9671**	**0.9553**	**0.9626**	**0.9885**	**0.9830**	**0.9869**	**0.9917**	**0.9837**	**0.9899**

TABLE VII
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON PU

Models	1%			5%			10%		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
SSUN(2018) [32]	0.8546	0.8768	0.8428	0.9359	0.9247	0.9151	0.9482	0.9362	0.9314
MCNN-CP(2020) [33]	0.8599	0.8205	0.8116	0.9395	0.9251	0.9197	0.9482	0.9413	0.9314
Tri-CNN(2023) [34]	0.9278	0.9081	0.9003	0.9413	0.9455	0.9537	0.9568	0.9627	0.9545
MDN-ASP(2021) [35]	0.9294	0.9082	0.9119	0.9538	0.9527	0.9414	0.9569	0.9678	0.9601
RIAN(2022) [36]	0.9358	0.9183	0.9248	0.9551	0.9602	0.9433	0.9608	0.9682	0.9614
DMuCA(2022) [37]	0.9364	0.9204	0.9219	0.9797	0.9751	0.9801	0.9709	0.9721	0.9633
MDGCN(2019) [38]	0.9378	0.9158	0.9267	0.9682	0.9517	0.9633	0.9768	0.9628	0.9712
Gia-CFSL(2022) [6]	0.9357	0.9214	0.9317	0.9725	0.9628	0.9733	0.9822	0.9725	0.9844
WFCG(2022) [39]	0.9456	0.9230	0.9345	0.9855	0.9729	0.9817	0.9934	0.9871	0.9914
GSC-CCTA	**0.9537**	**0.9242**	**0.9387**	**0.9879**	**0.9771**	**0.9833**	**0.9983**	**0.9881**	**0.9955**

TABLE VIII
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON SA

Models	1%			5%			10%		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
SSUN(2018) [32]	0.8959	0.9334	0.8841	0.9033	0.8823	0.8922	0.9195	0.8944	0.9102
MCNN-CP(2020) [33]	0.8862	0.9177	0.8731	0.9211	0.9462	0.9121	0.9369	0.9594	0.9286
Tri-CNN(2023) [34]	0.9271	0.8927	0.9043	0.9322	0.9288	0.9311	0.9536	0.9617	0.9526
MDN-ASP(2021) [35]	0.9366	0.9217	0.9204	0.9444	0.9547	0.9427	0.9552	0.9756	0.9625
RIAN(2022) [36]	0.9397	0.9254	0.9305	0.9546	0.9601	0.9488	0.9634	0.9759	0.9688
DMuCA(2022) [37]	0.9452	0.9367	0.9401	0.9587	0.9725	0.9614	0.9715	0.9801	0.9714
MDGCN(2019) [38]	0.9462	0.9498	0.9512	0.9673	0.9822	0.9636	0.9868	0.9817	0.9725
Gia-CFSL(2022) [6]	0.9662	0.9758	0.9699	0.9858	0.9816	0.9825	0.9963	0.9924	0.9826
WFCG(2022) [39]	0.9799	0.9817	0.9811	0.9895	0.9934	0.9914	0.9984	0.9988	0.9971
GSC-CCTA	**0.9877**	**0.9898**	**0.9863**	**0.9993**	**0.9991**	**0.9991**	**0.9995**	**0.9993**	**0.9997**

In addition, the confusion matrix given in Fig. 8 also shows that Class 16 for IP, Class 9 for PU, and Class 11 for Houston have the highest rate of being misclassified due to the fact that the three classes of ground objects, Buildings-Grass-Trees-Drives, Bitumen, and Lettuce-romaine-5wk, are dispersed and far away, which can lead to smoothing of the local information to obtain suboptimal results if the step size of the graph-structured convolution used for node access is too large.

1) Classification Ability on the IP Dataset: IP is an unbalanced dataset with scattered feature samples that make classification difficult in addition to the fact that the Oats and Grass-pasture-mowed categories of the IP dataset contain only 28 and 20 samples, respectively. To train the classifier, we need to select 5% of the training samples; the classification results for 10% and 15% training samples are also presented in this study. According to Table VI and Fig. 9, SSUN has the weakest classification performance and the most noise. This is due to the fact that it is a shallow neural network with insufficient generalization ability on limited sample data. Among the remaining eight methods, Gia-CFSL and WFCG have better classification performance than DMuCA and RIAN, which use graph convolutional networks to characterize the spatial dependencies of HSI data, indicating that the advantage of graph learning on HSI data is obvious. Meanwhile, the WFCG model that used graph

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TABLE IX
QUANTITATIVE RESULTS OF COMPARATIVE EXPERIMENTS ON HOUSTON

Models	0.1%			0.5%			1%		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
SSUN(2018) [32]	0.8125	0.8056	0.8231	0.9435	0.9520	0.9389	0.9845	0.9755	0.9829
MCNN-CP(2020) [33]	0.8269	0.8397	0.8178	0.9582	0.9627	0.9401	0.9817	0.9623	0.9779
Tri-CNN(2023) [34]	0.8296	0.8229	0.8355	0.9528	0.9436	0.9418	0.9625	0.9601	0.9734
MDN-ASP(2021) [35]	0.8299	0.8309	0.8367	0.9587	0.9527	0.9484	0.9719	0.9882	0.9752
RIAN(2022) [36]	0.8305	0.8311	0.8297	0.9627	0.9714	0.9732	0.9787	0.9799	0.9827
DMuCA(2022) [37]	0.8314	0.8323	0.8257	0.9605	0.9710	0.9741	0.9810	0.9757	0.9846
MDGCN(2019) [38]	0.8397	0.8257	0.8225	0.9617	0.9526	0.9596	0.9861	0.9859	0.9812
Gia-CFSL(2022) [6]	0.8405	0.8472	0.8322	0.9714	0.9703	0.9759	0.9901	0.9825	0.9834
WFCG(2022) [39]	0.8407	0.8510	0.8303	0.9643	0.9712	0.9628	0.9899	0.9903	0.9917
GSC-CCTA	**0.8551**	**0.8607**	**0.8433**	**0.9797**	**0.9751**	**0.9768**	**0.9907**	**0.9911**	**0.9941**

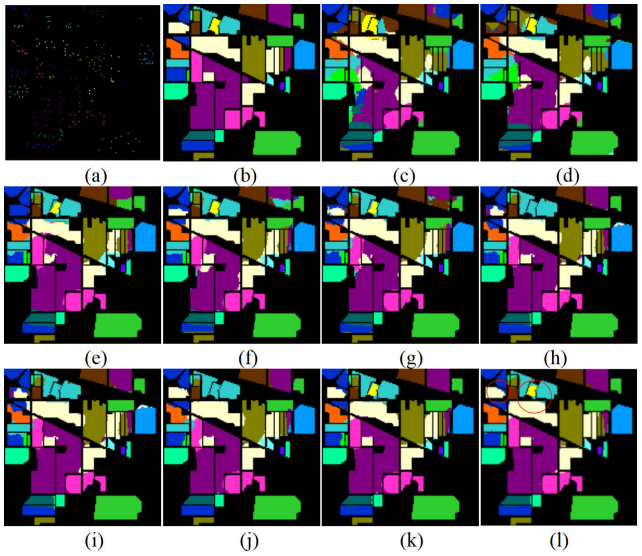


Fig. 9. Visualization results of comparative experiments on IP. (a) Input (5%). (b) GT. (c) SSUN. (d) MCNN-CP. (e) Tri-CNN. (f) MDN-ASP. (g) RIAN. (h) DMuCA. (i) MDGCN. (j) Gia-CFSL. (k) WFCG. (l) GSC-CCTA.

learning outperformed the Gia-CFSL that also used graph learning due to the introduction of the attention mechanism. In addition, the classification accuracy achieved with the attention model is much higher than SSUN and MCNN-CP because the trained attention mechanism tends to focus more on distinguishing characteristics. With cross-attention and structured graph convolution, GSC-CCTA achieves the best classification performance. Notably, MDGCN with subpar classification performance beats Tri-CNN due to the drive of the attention mechanism, demonstrating that the attention module can potentially boost the robustness of the HSIs classifier. Thus, it is clear that the superiority of graph learning and attention is demonstrated in the HSIs' classification task.

2) Classification Results on the PU Dataset: Complex backdrop pixels, a small amount of connected regions, and a big number of discontinuous pixels define the PU dataset. The experimental comparison findings indicate that the performance of each technique on this dataset is slightly enhanced. This may be because there are only nine classes in this dataset, and the spatial resolution is great, which decreases the classification complexity slightly. In the event of a very limited

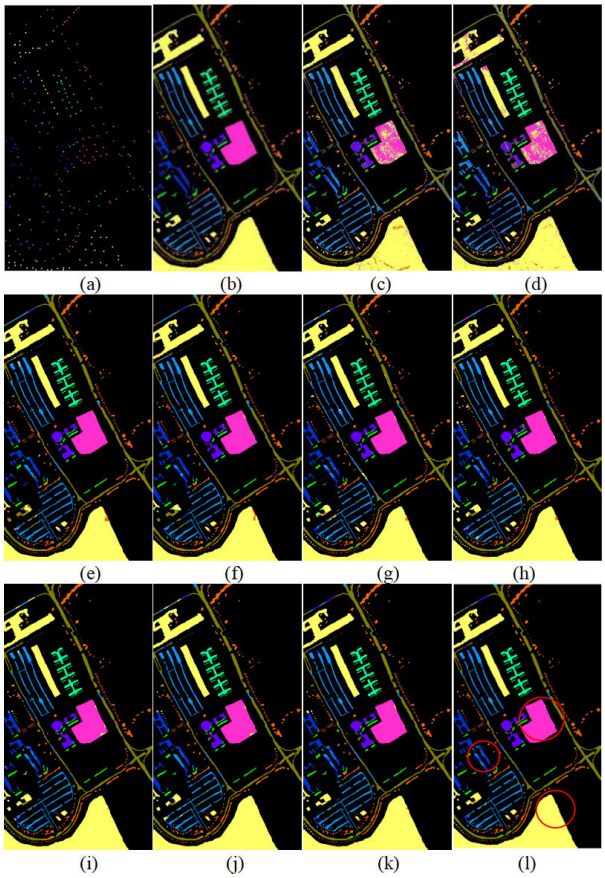


Fig. 10. Visualization results of comparative experiments on PU. (a) Input (1%). (b) GT. (c) SSUN. (d) MCNN-CP. (e) Tri-CNN. (f) MDN-ASP. (g) RIAN. (h) DMuCA. (i) MDGCN. (j) Gia-CFSL. (k) WFCG. (l) GSC-CCTA.

number of labeled data, the performance of the baseline technique remains inadequate. However, the proposed GSC-CCTA model achieved a competitive performance. The classification results achieved good classification results for discontinuous types of features compared to other algorithms. It can also be found from Table VII and Fig. 10 that, although Tri-CNN gets better OA than MCNN-CP and SSUN, MDN-ASP still outperforms Tri-CNN in two metrics, AA and Kappa, which indicates that the attention mechanism has stronger feature learning ability in small sample scenarios, while MDGCN

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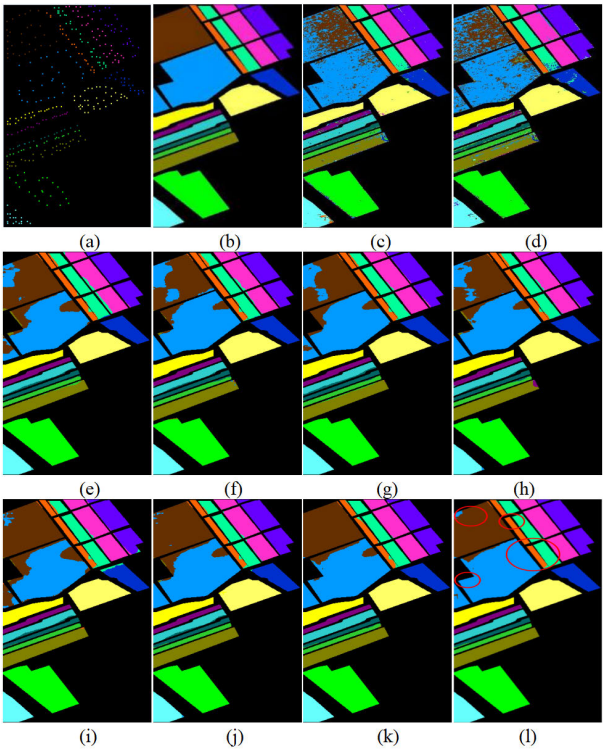


Fig. 11. Visualization results of comparative experiments on SA. (a) Input (1%). (b) GT. (c) SSUN. (d) MCNN-CP. (e) Tri-CNN. (f) MDN-ASP. (g) RIAN. (h) DMuCA. (i) MDGCN. (j) Gia-CFSL. (k) WFCG. (l) GSC-CCTA.

outperforms DMuCA to further validate the superiority of the attention mechanism on HSIs' classification tasks. Finally, the proposed GSC-CCTA provides state-of-the-art ability in all three metrics, with OA, AA, and Kappa improving by 0.84%, 0.12%, and 0.44% under 1% training samples.

3) Classification Results on the SA Dataset: As can be found from Table VIII and Fig. 11, all classifiers obtained high OA on the SA, but CNN- and attention-based methods did not classify the grapes_untrained class well, and only graph learning-based methods and the proposed GSC-CCTA methods obtained more robust results. This is because the traditional CNN-based methods have difficulty in effectively obtaining the spatial dependencies between spectra because of the influence of homospectral heterogeneity, so these three groups of methods based on graph learning provide superior classification results. In addition, the designed GSC-CCTA method improves the three metrics by 0.78%, 0.81%, and 0.52%, respectively.

4) Classification Results on the Houston Dataset: The Houston 2013 is larger and more complex than the three preceding datasets; therefore, we utilized only 0.1% of the samples for training. All approaches produce high OA on the Houston, as shown in Table IX and Fig. 12, although the fact that spectral properties influence the Stressed grass class and the performance of the other nine sets of approaches is inadequate. The proposed GSC-CCTA direction achieves the best classification performance by computing cross-attention to classify features by solving for the pinched cosine distance and employing a threshold-aware strategy to model multidimensional spatial relationships, which improves the representation of features in graph-structured data. The OA, AA, and Kappa coefficients at 0.1% training samples are, respectively, 1.68%, 1.12%, and 0.78% higher than the other approaches.

TABLE X
CLASSIFICATION RESULTS UNDER DIFFERENT THRESHOLDS

	Threshold					
	0.9	0.8	0.7	0.6	0.5	0.4
OA	0.9589	0.9601	**0.9671**	0.9605	0.9553	0.9422
AA	0.9408	0.9452	**0.9553**	0.9461	0.9410	0.9378
Kappa	0.9511	0.9517	**0.9626**	0.9489	0.9501	0.9452

TABLE XI
ABLATION EXPERIMENTS FOR THE SIZE OF THE SQUARE MOVING WINDOW

	IP (5%)				
	11×11	13×13	15×15	17×17	19×19
OA	0.9555	0.9636	**0.9671**	0.9656	0.9606
AA	0.9518	0.9551	**0.9553**	0.9521	0.9523
Kappa	0.9494	0.9586	**0.9626**	0.9608	0.9553

E. Parameter Sensitivity Analysis

In this article, the sensitivity of cross-attention to more discriminative features is enhanced by introducing a threshold T to control the distance of the angle cosine. To determine the optimal T value, sensitivity analysis experiments are conducted in this section on the IP dataset using 5% training samples. According to Table X, as the threshold value decreases, the classification ability of GSC-CCTA increases and peaks when the threshold value is 0.7, after which the classification ability decreases as the threshold value continues to decrease; therefore, 0.7 is the optimal threshold value chosen in this article. Meanwhile, it also shows that the larger the angle cosine is, the more effective features of HSIs can be extracted.

In addition, Table XI also gives the sensitivity analysis results of the size of the square moving window of the first-order statistical operator, the results show that the classification ability rises with the increase of the size of the moving window, and the optimal classification performance is achieved when the size of the moving window is 15 × 15 and then shows a decreasing tendency so that the patch features generated when the size of the moving window is 15 × 15 symbolize the more discriminative spatial–spectral information.

F. Performance Trends of Different Training Samples

This section focuses on training robust classifiers on few samples, but, to thoroughly test the efficacy of the designed GSC-CCTA, it uses 0.1%–60% samples to train on the IP, and the test data are displayed in Fig. 13. Despite the fact that only 5% of the training data is utilized by the proposed method on IP, the OA is still as high as 96.71%. The performance curves on the four datasets clearly illustrate the sharp increase

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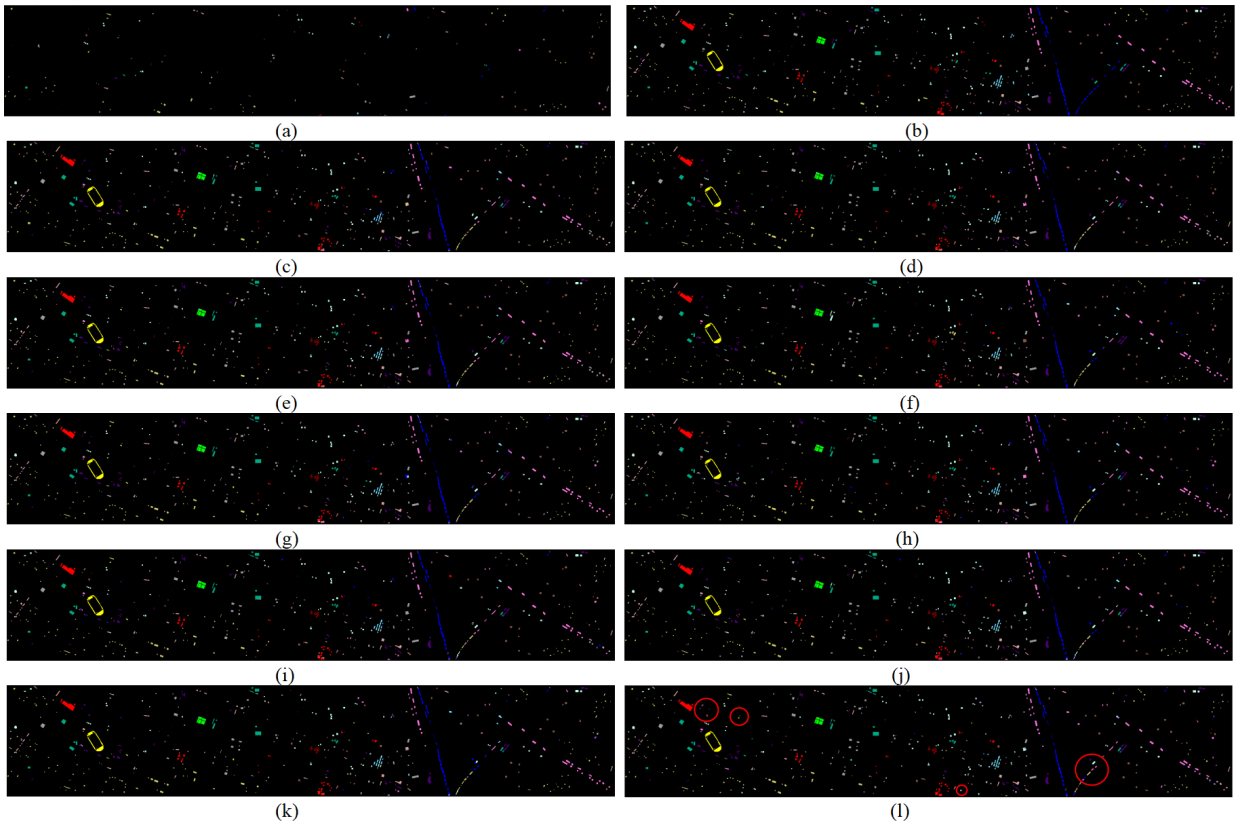


Fig. 12. Visualization results of comparative experiments on Houston. (a) Input (0.1%). (b) GT. (c) SSUN. (d) MCNN-CP. (e) Tri-CNN. (f) MDN-ASP. (g) RIAN. (h) DMuCA. (i) MDGCN. (j) Gia-CFSL. (k) WFCG. (l) GSC-CCTA.

TABLE XII
TEST RESULTS OF THE CROSS-ATTENTION ABLATION

	IP(5%)			PU(1%)			SA(1%)			Houston(0.1%)		
	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA
V	0.9480	0.9543	0.9248	0.9256	0.9440	0.9098	0.9748	0.9773	0.9889	0.8157	0.8297	0.8292
H	0.9517	0.9576	0.9513	0.9326	0.9491	0.9157	0.9789	0.9810	0.9890	0.8278	0.8409	0.8364
H-V(Ours)	**0.9626**	**0.9671**	**0.9553**	**0.9387**	**0.9537**	**0.9242**	**0.9863**	**0.9877**	**0.9898**	**0.8433**	**0.8551**	**0.8607**

in classification ability as the number of training samples grows, and all obtain close to 100% OA when using 30% of the samples, which fully demonstrates the superiority of GSC-CCTA in a limited sample environment.

G. Ablation Experiments for Cross-Attention Module

This section runs an ablation test to determine the plausibility of the developed cross-attention module. On the Houston dataset, ablation tests are conducted using 0.1% training samples. The findings of the ablation experiment are presented in Table XII with "V" representing vertical attention, "H" representing horizontal attention, and "H-V" representing cross-attention.

As indicated in Table XII and Fig. 14, the test results of horizontal unidirectional attention are better than vertical unidirectional attention, but cross-attention is significantly better than unidirectional attention. Therefore, the proposed cross-attention weights obtained from two directions are effective.

TABLE XIII
TEST RESULTS OF THE SUBMODULES' ABLATION

Type	IP(5%)		
	Kappa	OA	AA
NO-CCTA	0.9489	0.9516	0.9412
NO-GSC	0.9367	0.9487	0.9369
GSC-CCTA	**0.9626**	**0.9671**	**0.9553**

H. Ablation Experiment for Submodules

To verify the effectiveness of GSC-CCTA, this section investigates the ablation of its submodules. Let "NO-GSC" mean using CNN instead of graph-structured convolution and "NO-CCTA" mean not using a continuous content awareness module. Table XIII demonstrates that when GSC is not used, the classification ability reduces by 1.26%, 1.02%, and 1.97%, respectively. When CCTA is not used, the classification performance decreases by 0.96%, 0.57%, and 0.7%, respectively. This indicates that GSC is more important than CCTA, which also illustrates the superiority of graph-structured convolution for hyperspectral classification tasks.

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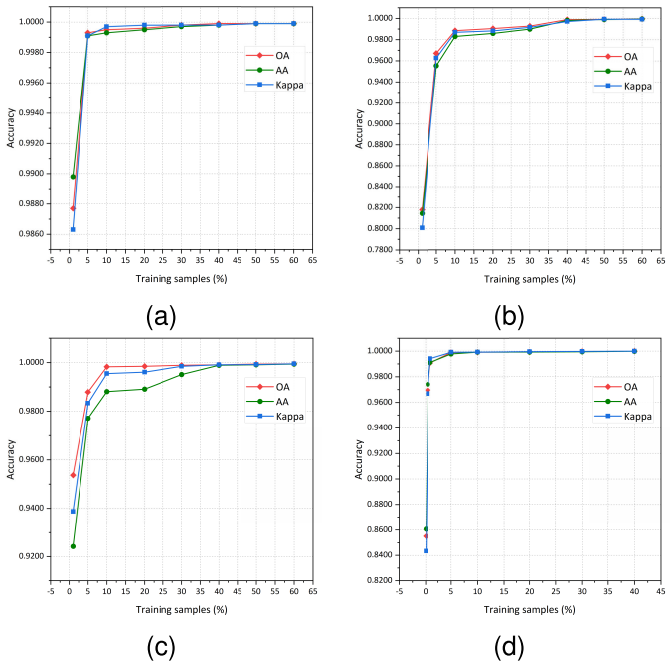


Fig. 13. Performance trends of the GSC-CCTA with increasing training samples. (a) SA. (b) IP. (c) PU. (d) Houston.

TABLE XIV
RESULTS OF ABLATION EXPERIMENTS FOR STEP SIZE OF NODE VISIT

	k (IP:5%)				
	2	3	4	5	6
OA	0.9215	0.9574	**0.9671**	0.9310	0.8962
AA	0.9178	0.9632	**0.9553**	0.9271	0.8829
Kappa	0.9203	0.9514	**0.9626**	0.9226	0.8893

I. Ablation Experiments for Step Size of Node Visit

To obtain the optimal step size for node visits, the ablation studies are carried out in this section.

As can be seen from Table XIV, the classification ability of the proposed algorithm is degraded regardless of whether the step size k is too small, or too large, and furthermore, the classification ability degrades even more drastically if the step size is too large. Therefore, although we have determined that a step size of 4 results in poor classification accuracy for very few classes, overall, the proposed method is effective and superior.

J. t-SNE Data Distributions Visualization

To confirm the superiority of the proposed methodology further, we employ t-distributed stochastic neighbor embedding (t-SNE) to visualize the predicted output of the model on the IP. The visualization results of GSC-CCTA and the other nine groups of classifiers compared are given in Fig. 15, from which it is evident that the designed GSC-CCTA gets the best classification result. Moreover, SSUN, MCNN-CP, and Tri-CNN have lower clustering accuracy, while MDGCN, Gia-CFSL, and WFCG achieve optimal results with more compact intraclasses and more separation between classes. This is because graph convolution mines more discriminative directional features. Of course, the superiority of GSC-CCTA is mainly attributed to the proposed graph-structured convolution and continuous threshold-aware modules.

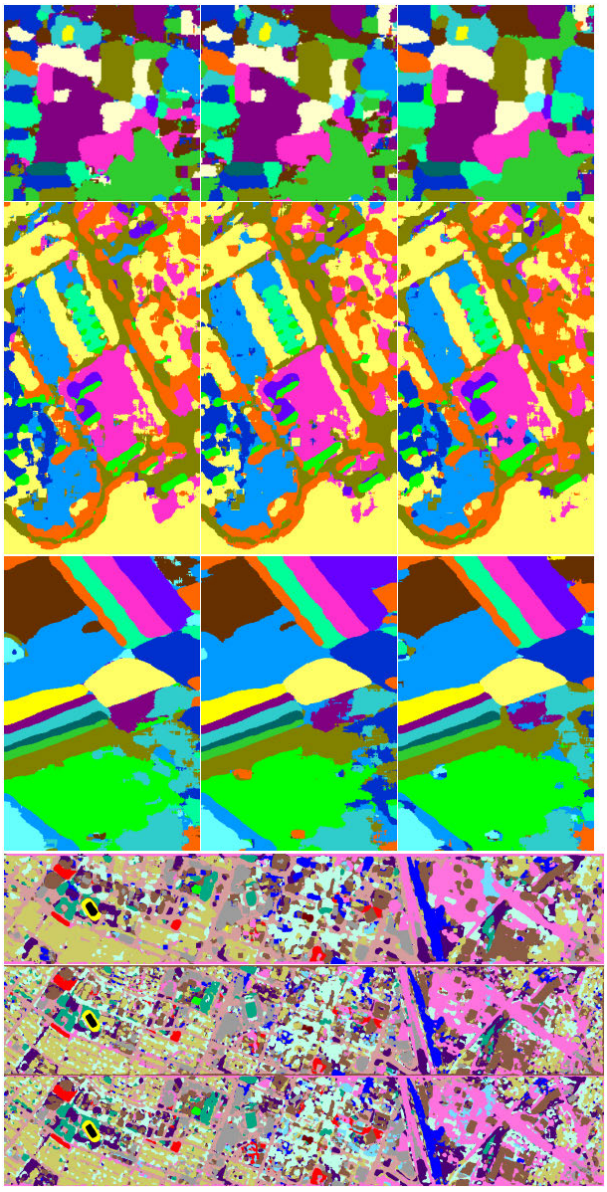


Fig. 14. From (left) to (right) [Houston is from (top) to (bottom)] are the full classification results using vertical, horizontal, and cross-attention, where IP, PU, SA, and Houston are using 5%, 1%, 1%, and 0.1% of the training samples, respectively.

V. CONCLUSION

In this article, we propose a novel graph-structured convolution-guided continuous context threshold-aware module for HSIs' classification of low training samples, which models the spectral–spatial vectors using second- and first-order pooling operators, respectively, which outperforms traditional dimensionality reduction methods and preserves the spatial correlation and contextual information of the spectra. In addition, utilizing the random walk transition probability matrix, the graph-structured convolution can mine more discriminative directional features, and our improved graph-structured convolution preserves the negative part of the deep information. In addition, the designed continuous

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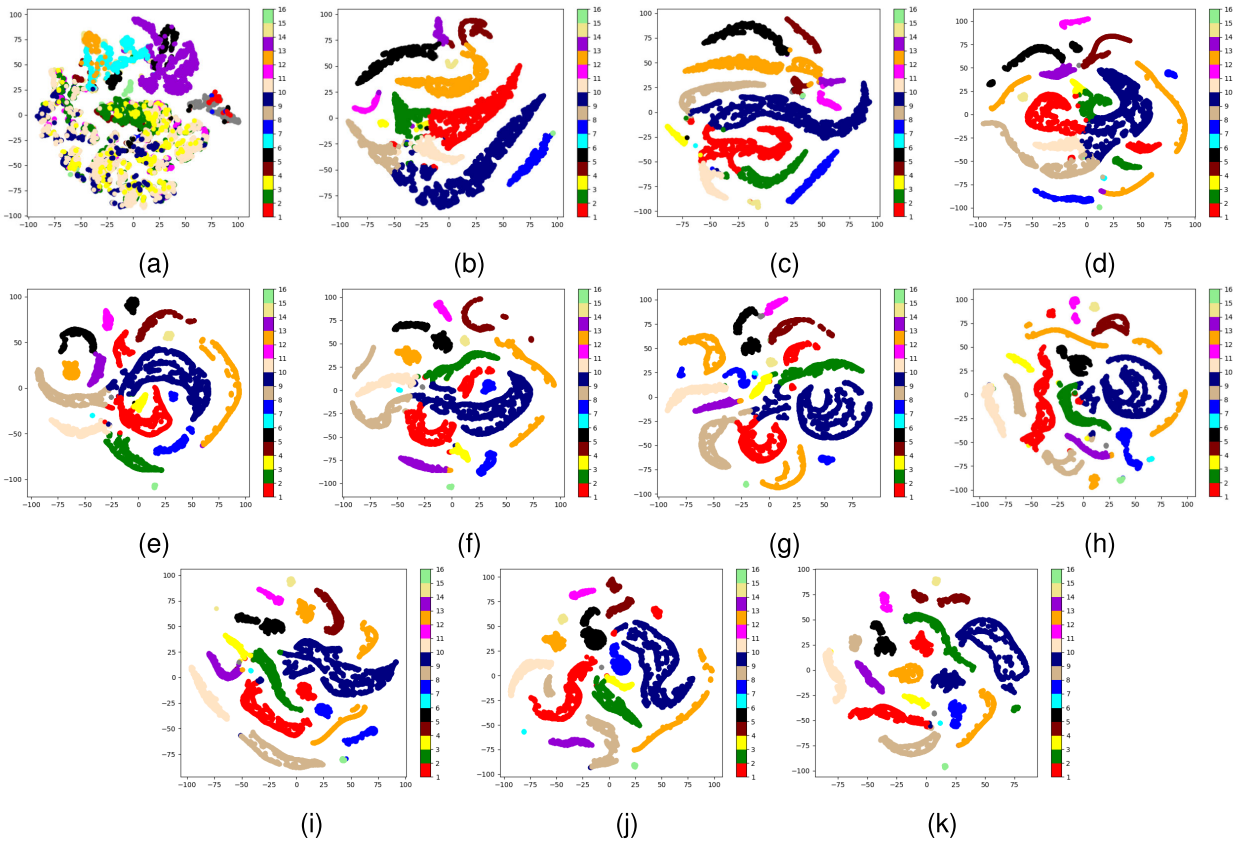


Fig. 15. t-SNE feature visualization of GSC-CCTA and the other nine comparison models. (a) Raw. (b) SSUN. (c) MCNN-CP. (d) Tri-CNN. (e) MDN-ASP. (f) RIAN. (g) DMuCA. (h) MDGCN. (i) Gia-CFSL. (j) WFCG. (k) GSC-CCTA.

context threshold-aware network is able to mine multidimensional spatial relations, which enhances the representation ability of graph-structured convolution. Through ablation and comparison tests, not only the effectiveness of the improved graph-structured convolution and threshold-aware network is demonstrated but also the superiority of the cross-attention mechanism in controlling the feature filtering through the angular cosine distance, which significantly improves the classification ability of the proposed method. Extensive tests on the four most commonly used HSI datasets demonstrate the competitive performance of the proposed method. Moreover, since the step size of the graph-structured convolution for node access should not be too large, suboptimal results will be achieved for this class if the ground objects are scattered or far away. Therefore, in future work, we will focus on this limitation to reduce local information being smoothed while increasing the step size.

REFERENCES

[1] A. F. Gao, B. Rasmussen, P. Kulits, E. L. Scheller, R. Greenberger, and B. L. Ehlmann, "Generalized unsupervised clustering of hyperspectral images of geological targets in the near infrared," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 4294–4303.

[2] P. J. Zarco-Tejada, M. V. González-Dugo, and E. Fereres, "Seasonal stability of chlorophyll fluorescence quantified from airborne hyperspectral imagery as an indicator of net photosynthesis in the context of precision agriculture," *Remote Sens. Environ.*, vol. 179, pp. 89–103, Jun. 2016.

[3] G. Sun, Z. Jiao, A. Zhang, F. Li, H. Fu, and Z. Li, "Hyperspectral image-based vegetation index (HSVI): A new vegetation index for urban ecological research," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102529.

[4] M. Zhang, W. Li, Q. Du, L. Gao, and B. Zhang, "Feature extraction for classification of hyperspectral and LiDAR data using patch-to-patch CNN," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 100–111, Jan. 2020.

[5] Y. Dong, W. Shi, B. Du, X. Hu, and L. Zhang, "Asymmetric weighted logistic metric learning for hyperspectral target detection," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 11093–11106, Oct. 2022.

[6] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 30, 2022, doi: 10.1109/TNNLS.2022.3185795.

[7] D. Hong, N. Yokoya, J. Chanussot, J. Xu, and X. X. Zhu, "Joint and progressive subspace analysis (JPSA) with spatial–spectral manifold alignment for semisupervised hyperspectral dimensionality reduction," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3602–3615, Jul. 2021.

[8] Z. Xue, M. Zhang, Y. Liu, and P. Du, "Attention-based second-order pooling network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9600–9615, Nov. 2021.

[9] R. Shang, H. Chang, W. Zhang, J. Feng, Y. Li, and L. Jiao, "Hyperspectral image classification based on multiscale cross-branch response and second-order channel attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532016.

[10] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-D CNN for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2020.3043710.

[11] Y. Zhang, G. Cao, B. Wang, X. Li, P. Y. Owusu Amoako, and A. Shafique, "Dual sparse representation graph-based copropagation for semisupervised hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500517, doi: 10.1109/TGRS.2020.3046780.

[12] Y. Kong, X. Wang, Y. Cheng, Y. Chen, and C. L. P. Chen, "Graph domain adversarial network with dual-weighted pseudo-label loss for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3135310.

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[13] J. Jiang, J. Ma, and X. Liu, "Multilayer spectral–spatial graphs for label noisy robust hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 839–852, Feb. 2022, doi: 10.1109/TNNLS.2020.3029523.

[14] S. Dong, W. Feng, Y. Quan, G. Dauphin, L. Gao, and M. Xing, "Deep ensemble CNN method based on sample expansion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531815, doi: 10.1109/TGRS.2022.3183189.

[15] Y. Xu, B. Du, and L. Zhang, "Beyond the patchwise classification: Spectral–spatial fully convolutional networks for hyperspectral image classification," *IEEE Trans. Big Data*, vol. 6, no. 3, pp. 492–506, Sep. 2020.

[16] Y. Hechtlinger, P. Chakravarti, and J. Qin, "A generalization of convolutional neural networks to graph-structured data," 2017, *arXiv:1704.08165*.

[17] X. Zhou et al., "Graph convolutional network hashing," *IEEE Trans. Cybern.*, vol. 50, no. 4, pp. 1460–1472, Apr. 2020.

[18] M. M. Bronstein, J. Bruna, Y. LeCun, A. Szlam, and P. Vandergheynst, "Geometric deep learning: Going beyond Euclidean data," *IEEE Signal Process. Mag.*, vol. 34, no. 4, pp. 18–42, Jul. 2017.

[19] Y. Ma, J. Hao, Y. Yang, H. Li, J. Jin, and G. Chen, "Spectral-based graph convolutional network for directed graphs," 2019, *arXiv:1907.08990*.

[20] M. Henaff, J. Bruna, and Y. LeCun, "Deep convolutional networks on graph-structured data," 2015, *arXiv:1506.05163*.

[21] S. Kumar, J. Ying, J. V. D. M. Cardoso, and D. Palomar, "Structured graph learning via Laplacian spectral constraints," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–13.

[22] J. Atwood and D. Towsley, "Diffusion-convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 1–9.

[23] L. Bai, L. Cui, Y. Jiao, L. Rossi, and E. R. Hancock, "Learning backtrackless aligned-spatial graph convolutional networks for graph classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 2, pp. 783–798, Feb. 2022.

[24] T. Danel et al., "Spatial graph convolutional networks," in *Proc. Int. Conf. Neural Inf. Process.* Cham, Switzerland: Springer, Nov. 2020, pp. 668–675.

[25] F. Luo, B. Du, L. Zhang, L. Zhang, and D. Tao, "Feature learning using spatial–spectral hypergraph discriminant analysis for hyperspectral image," *IEEE Trans. Cybern.*, vol. 49, no. 7, pp. 2406–2419, Jul. 2019, doi: 10.1109/TCYB.2018.2810806.

[26] Y. Yang et al., "Semi-supervised multiscale dynamic graph convolution network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Oct. 21, 2022, doi: 10.1109/TNNLS.2022.3212985.

[27] R. Luo, W. Liao, X. Huang, Y. Pi, and W. Philips, "Feature extraction of hyperspectral images with semisupervised graph learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4389–4399, Sep. 2016.

[28] P. Sellars, A. I. Aviles-Rivero, and C.-B. Schönlieb, "Superpixel contracted graph-based learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4180–4193, Jun. 2020.

[29] S. Wan et al., "Multi-level graph learning network for hyperspectral image classification," *Pattern Recognit.*, vol. 129, Sep. 2022, Art. no. 108705.

[30] Y. Ding, X. Zhao, Z. Zhang, W. Cai, and N. Yang, "Graph sample and aggregate-attention network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] K. Xu, Y. Zhao, L. Zhang, C. Gao, and H. Huang, "Spectral–spatial residual graph attention network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[32] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral–spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.

[33] J. Zheng, Y. Feng, C. Bai, and J. Zhang, "Hyperspectral image classification using mixed convolutions and covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 522–534, Jan. 2021.

[34] M. Q. Alkhatib et al., "Tri-CNN: A three branch model for hyperspectral image classification," *Remote Sens.*, vol. 15, no. 2, p. 316, Jan. 2023.

[35] B. Xi et al., "Multi-direction networks with attentional spectral prior for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500915.

[36] X. Zheng, H. Sun, X. Lu, and W. Xie, "Rotation-invariant attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 4251–4265, 2022, doi: 10.1109/TIP.2022.3177322.

[37] M. Liang, Q. He, X. Yu, H. Wang, Z. Meng, and L. Jiao, "A dual multi-head contextual attention network for hyperspectral image classification," *Remote Sens.*, vol. 14, no. 13, p. 3091, Jun. 2022.

[38] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.

[39] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022, doi: 10.1109/TIP.2022.3144017.

[40] Y. Gao, R. Ji, P. Cui, Q. Dai, and G. Hua, "Hyperspectral image classification through bilayer graph-based learning," *IEEE Trans. Image Process.*, vol. 23, no. 7, pp. 2769–2778, Jul. 2014, doi: 10.1109/TIP.2014.2319735.

[41] D. J. Im and G. W. Taylor, "Semisupervised hyperspectral image classification via neighborhood graph learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 9, pp. 1913–1917, Sep. 2015, doi: 10.1109/LGRS.2015.2438227.

[42] L. Ma, M. M. Crawford, X. Yang, and Y. Guo, "Local-manifold-learning-based graph construction for semisupervised hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2832–2844, May 2015, doi: 10.1109/TGRS.2014.2365676.

[43] Z. Xue, P. Du, J. Li, and H. Su, "Simultaneous sparse graph embedding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 11, pp. 6114–6133, Nov. 2015, doi: 10.1109/TGRS.2015.2432059.

[44] Z. Xue, S. Yang, and L. Zhang, "Weighted sparse graph regularization for spectral–spatial classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 9, pp. 1630–1634, Sep. 2021, doi: 10.1109/LGRS.2020.3005168.

[45] Y. Su, M. Jiang, L. Gao, X. Sun, X. You, and P. Li, "Graph-cut-based collaborative node embeddings for hyperspectral images classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2022.3184817.

[46] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2022, doi: 10.1109/TCYB.2022.3219855.

[47] W. Guo, G. Xu, W. Liu, B. Liu, and Y. Wang, "CNN-combined graph residual network with multilevel feature fusion for hyperspectral image classification," *IET Comput. Vis.*, vol. 15, no. 8, pp. 592–607, Dec. 2021.

[48] M. Lin, W. Jing, D. Di, G. Chen, and H. Song, "Context-aware attentional graph U-Net for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3069987.

[49] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021, doi: 10.1109/TGRS.2020.3037361.

[50] L. Gao, X. Sun, X. Sun, L. Zhuang, Q. Du, and B. Zhang, "Hyperspectral anomaly detection based on chessboard topology," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505016, doi: 10.1109/TGRS.2023.3249748.



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